

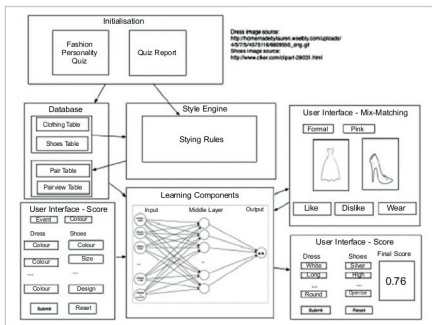


Fig. 5: Schematic overview of Smart Fashion architecture

There are four popular applications of an AI-based stylist program:

1. Internet-based human stylist consultant services for putting communication between clients and stylists on the Internet. It helps to improve the flexibility and accessibility of styling work.
2. Virtual fitting systems to fill one of the major gaps between e-commerce and retail stores. It enables customers to go through the product line, watch over product reviews and gain more confidence about the item being purchased.
3. Recommender system implementations in e-commerce to recommend new items based on users' browsing histories.
4. Online fashion communities to provide a platform for people to share, gain and communicate fashion inspiration and shopping information online.

How AI helps in fashion design

Garment representation. As mentioned earlier, a set of basic features such as colour, shape, print and fabric is used to describe garments. Computer vision techniques automatically recognise the first three features. Colour is represented in RGB (red, green and blue) format, which is later

converted into HSI (hue, saturation and intensity) model.

Shape is easily extracted using computer vision techniques and print of a garment is considered in its loudness. Loudness is defined by the frequency of colour changes in the garment and also by changes in placement on the garment.

Fabric is the most complex aspect as even humans have trouble in recognising a fabric just by looking at it. One

possible solution for tricky attributes like fabric is stylistic semantic correlations between clothing attributes. Systems learn correlations between attributes and then intelligently fill in missing values. For example, for a formal dress there is a high possibility of it being made of silk, whereas casual wear such as t-shirts are more likely to be made of cotton.

Support vector machine classifiers are applied on all attributes to determine how useful these attributes would be in prediction. Then, the system makes predictions based on the inference of different attributes and mutual dependency relations.

Colour harmony evaluation. Matsuda Colour Coordination (MCC) developed by Yutaka Matsuda is used to evaluate the harmony rate between colours. A total of 80 colour schemes (8 hue types and 10 tone types) are used. A colour scheme is considered harmonious if there are many samples in the system.

Each individual has a unique style preference. An intelligent style system can instantaneously adapt from a standard colour scheme to suit a user's personal preference. Linguistic labels (neutral, a little, slightly, fairly, very and extremely) are added on the colour scheme and presented as fuzzy sets, respectively. The system adapts to a user's preference during its inter-

User questionnaire

1. What is your height?
A. Short; B. Medium; C. Tall
2. How will you describe your body figure?
A. Balanced; B. Sinewy; C. Hourglass; D. Broad; E. Stocky
3. Describe your face feature?
A. Even; B. Angular; C. Rounded; D. Youthful; E. Sharp
4. What is your hair style?
A. Moderate to short in length, well groomed, controlled & smooth;
B. Long with curves and curls;
C. Short with feather cut around the face;
D. Soft and bouncy, and never straight;
E. "Gibson Girl" style;
F. Sleek, geometric or asymmetrical cuts;
G. Tousled, loose, windblown;
H. Short cropped boyish cuts.
5. How will you describe your shoulder and hips?
A. Evenly proportioned; B. Curvy with rounded shoulder; C. Inverted triangle
6. What your eyes look like?
A. Evenly shaped; B. Angular and pointed; C. Large; D. Soft and round
7. What your lips look like?
A. Moderate, evenly shaped; B. Thin, angular and delicate; C. Full, luscious and thick
8. What your feet look like?
A. Small; B. Average; C. Large

(Source: www.ai.uga.edu)